TECHNOLOGY, INDUSTRIALISATION AND DEVELOPMENT (TIDE) CENTRE

WORKING PAPER 091

# Can AI grow green?

## Evidence of an inverted-U curve between AI, energy use and emissions

Angel Melguizo, Raúl Katz & Juan Jung

JUNE 2025

## ABOUT THE TIDE CENTRE

The Technology and Industrialisation for Development Centre is as a multidisciplinary research hub within the University of Oxford's Department of International Development. Founded in 2008 by Professor Xiaolan Fu, which brought together scholars and practitioners who view technology and industrialisation as essential drivers of sustainable development, the centre explores the complex relationships between trade, investment, technology, innovation and inclusive prosperity.

The Centre (formerly known as TMCD) has built on the pioneering work since the 1970s by Oxford's influential development economists — Frances Stewart, Sanjaya Lall, Barbara Harriss-White, Adrian Wood, and Valpy FitzGerald. In an era when the development agenda was dominated by concerns around food security, poverty reduction, and aid, these thoughts leaders have championed the transformative power of technology and industrialisation as key levers for progress.

In 2024, TMCD rebranded as the Technology and Industrialisation for Development Centre (TIDE) and appointed a new director, Amir Lebdioui. Today, the centre's mission is more crucial than ever. With increasing disparities in technological capabilities and a renewed global industrialization race, especially in the transition to a low-carbon economy, it is imperative to push the knowledge frontiers on innovation and industrial policy for local prosperity and true sustainability for all.

Find more about us at oxford-tide.org

# Can AI grow green?

## *Evidence of an inverted-U curve between AI, energy use and emissions*

**Ángel Melguizo**, ARGIA Green, Tech & Economics and TIDE Centre, University of Oxford

**Raúl Katz,** Columbia Institute for Tele-Information, Columbia University

**Juan Jung,** Faculty of Economics and Business, Pontifical Comillas University

**Abstract**

The relationship between artificial intelligence (AI) and the *green agenda* is one of the key current global economic and social topics, driven by conflicting assumptions and evidence. On the one hand, AI use cases can drive energy savings, support renewable energy transition, and reduce emissions. However, its initial adoption significantly increases energy consumption, thereby deepening the challenges countries and organizations face to reach their environmental sustainability goals. This paper presents novel empirical evidence on AI development and its environmental implications in 23 middle and high-income countries, confirming that, initially, in the majority of countries AI increases energy consumption and CO2 emissions. However, we also show that these relationships are not linear, since, over time, AI has a positive impact on the environment in terms of emission reduction and higher reliance on renewable energies, a kind of *Green AI Kuznets curve*. This reversal in the trend is achieved from $220-$580 AI market per capita. As of today, AI leading countries, such as Singapore and the US, are already benefitting from this *technological dividend*. These results have clear policy implications, calling for a less fragmented global AI and energy governance, national AI strategies, innovations in financing to mobilize additional funds towards greener AI adoption, while achieving more transparency and standards for measuring and reporting its use of energy.

Keywords: AI, energy, sustainability
JEL Classification: O33, Q55, L86

---

The authors would like to thank Alejandra Pallás Villacampa (Universidad Pontificia Comillas) for her research assistance, and the excellent comments from TIDE Centre, -University of Oxford colleagues Sabrina Fernandes, Tin Hinane el Kadi, Amir Lebdioui and Riad Meddeb. The authors relied on Perplexity PRO AI tool to map all existing AI spending databases and Elicit to assist in the review of the research literature. No AI tool was used for drafting or conducting econometric work. Email for comments amelguizo@argiagte.com

## 1. Introduction

Artificial intelligence (AI) is driving radical changes in production and consumer welfare. Generative AI could add up to US$ 4.4 trillion to the global economy, a positive shock not just for productivity in the technology and telecommunications sectors, but also in finance, retail and the life sciences (McKinsey, 2023). These remarkable effects stem both from labor productivity and automation, and an acceleration of innovation in a scenario where AI, as a general-purpose technology, spreads across the economy (Baily et al., 2024). Many caveats in this assessment are in order. First, AI's full potential remains untapped. Some authors argue that at least in the next decade AI impact will only involve automation of discrete tasks and some task complementarity, with modest impact on productivity, while others are more optimistic (for contrasting economic perspectives see Acemoglu, 2024 vs Baily et al., 2024; and OECD, 2024 for a survey perspective confirming the wide potential impact estimates). Second, in the absence of a more integrated global governance – particularly between low- and middle-income countries and high-income ones - and under the existing significant human capital and technology gaps across and within countries, AI development can reproduce and even amplify socioeconomic and environmental risks.

Focusing on the environmental risks, one of the most pressing issues raised by the AI boom is the significant resource consumption associated with AI innovation, from minerals and rare earth extraction for manufacturing the required semiconductors and batteries, to water and energy required by data centers' operations in the training of large language models (LLMs) (Lebdioui et al., 2025). Reporting on this extraordinary surge in energy demand from AI development and data centers' consumption of electricity and water, and their contribution to greenhouse effects and climate change appear regularly even in the general media (Crownhart, 2024; Mehta, 2024). Companies' environmental reports, in fact, confirm a significant increase in electricity and natural resource consumption, partially due to the AI boom. Greenhouse gas emissions from three top AI platform suppliers - Alphabet, Amazon and Microsoft - are up 62 percent from 2020, reaching 47 million metric tons alone in 2023 (ITU and WBA, 2024), which amounts to half of the total emissions of a country like Peru. Electricity use by these three companies has grown even faster, up 78 percent and standing at just over 100 TWh in 2023, similar to Colombia and Dominican Republic joint annual consumption. The latest *quasi-official* projections from the International Energy Agency (IEA, 2025) project that data center electricity consumption is set to more than double to around 945 TWh by 2030, slightly more than Japan's total electricity consumption today.

However, admittedly, a challenge for evaluating the relationship between AI, energy and biodiversity is the lack of emissions data, as prominent model developers such as OpenAI, Google, Anthropic or Mistral do not report emissions in training LLMs, a particularly high energy-consumption phase given that it involves numerous resource-intensive iterations (Stanford Institute for Human-Centered AI, 2024, and ITU, 2024). External reports show that currently, a single OpenAI´s ChatGPT search consumes as ten Google searches (Lokwon, 2025). Partially, due to the lack of reliable data, the empirical academic research on the impact of AI on energy consumption and on the environment is still relatively thin, and fragmented, lacking consistent measurement methodologies.

This paper aims to contribute towards filling this gap, providing empirical evidence on the relationship between AI investment, energy requirements, CO2 emissions and the energy transition towards renewables. For this purpose, we have built an original database covering 23 countries – including all AI-leaders worldwide, and tested these relations, with a focus on non-linearities to capture the potential effect of AI after it is widely adopted and/or consolidated. This can be seen as a test for a *Kuznets curve* for green AI. We find that AI initially has a negative impact on environmental indicators but turns a positive effect once a spending threshold is crossed. We believe this to be the first contribution providing evidence on the inverted-U relationship among the variables mentioned above, covering all AI leaders (both users and producers) during the past five years.

The paper is structured as follows. Section 2 reviews the research literature and sets the main empirical hypotheses. Section 3 described the data, while results and policy implications are reported in Section 4. Section 5 concludes and discusses limitations and potential research extensions of the paper.

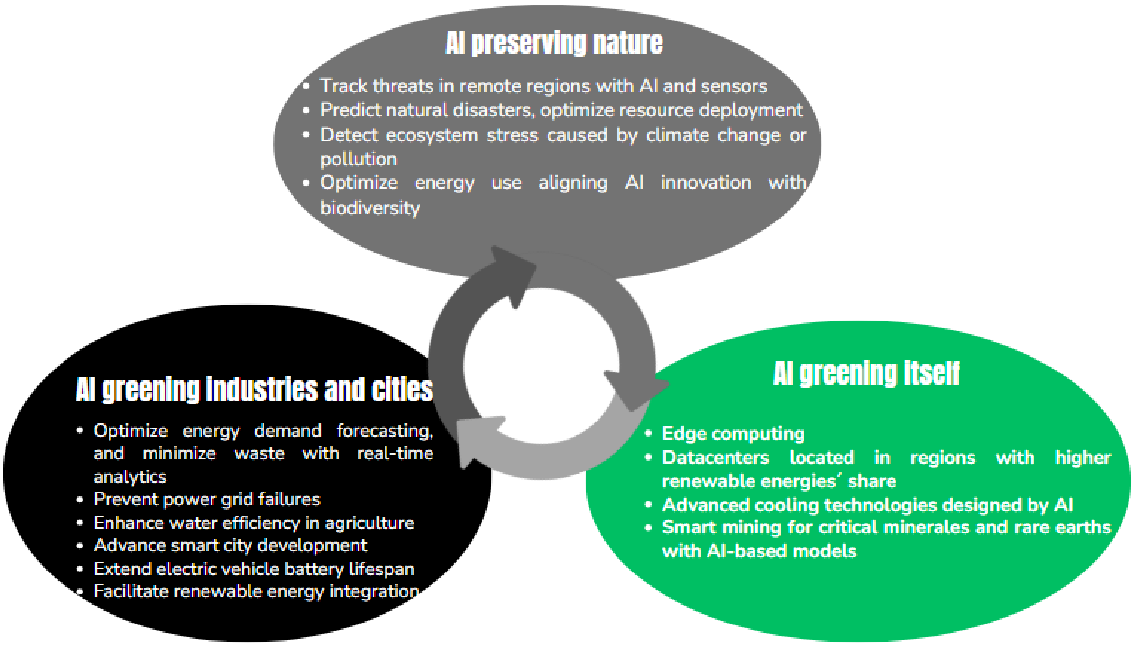
## 2. Literature review and hypotheses

Despite the recent surge of AI as a general-purpose technology, empirical academic research on AI energy consumption is still relatively limited due to the lack of available data. That said, emerging findings confirm a tension between AI development, resource use (e.g. water and energy) and environmental sustainability.

For starters, <u>AI algorithms are energy-intensive, leading to increased CO2 emissions</u>. A seminal study covering some of the most popular LLMs in 2022 (i.e. before the generative AI boom) showed that the training of BLOOM demanded energy equivalent to 25 times that of flying one passenger round trip from New York to San Francisco (Stanford Institute for Human-Centered AI, 2023; see Luccioni et al., 2024 for estimated CO2 emissions of different AI models performing different tasks). Even with lighter and more efficient models as DeepSeek´s, AI energy demand is projected to increase (according to the so-called *Jevons paradox,* as AI systems become more efficient and accessible, their usage and the resources they consume increase due to wider adoption and increased demand, e.g. Rosalsky, 2025).

On the positive side, AI applications are becoming a key component of the so-called *green agenda* (Figure 1).[1] AI can optimize and potentially reduce energy consumption and CO2 emissions. AI can contribute *to greening the economy*. Widespread adoption of existing AI applications could save around 8 exajoules (EJ) of industry energy demand by 2035, equivalent to more than the total energy demand of Mexico today, according to the IEA (2025). For example, in agriculture, AI-driven precision farming reduces energy consumption while enhancing productivity, by employing sensors and advanced analytics to monitor soil, water and climate. Similarly, AI implementation in manufacturing enhances resource efficiency through real-time shop-floor monitoring enabled by IoT and 5G networks (Tace et al., 2023). In the energy sector AI can significantly improve efficiency through predictive analysis, optimizing real-time energy consumption and adjusting demand based on environmental and operational data, as seen in applications like smart grids and energy management systems. AI further enhances flexibility by forecasting supply and demand, optimizing production patterns based on historical data, environmental conditions and power plant capacities, thus minimizing energy waste (Rojek et al., 2023, Rozite et al., 2023, Serebryantseva, 2023, and Şerban and Lytras, 2020).[2]

Figure 1. **AI and the *green agenda***




*Source:* Lebdioui, Melguizo and Muñoz (2025)

[1] This *AI and green agenda* framework is based on Lebdioui et al. (2025). We refer the interested readers to it, including the AI and biodiversity preservation angle not developed in this paper.
[2] An additional increasingly relevant dimension is that of AI and security, given energy growing geopolitical role and its advanced digitization (see for instance Lewis and Oxby, 2024). This is a natural extension of this research.

In addition, AI is launching initiatives to *"green" itself,* thanks to advanced cooling technologies and dynamic workload management. Also, edge computing reduces energy demand by processing data closer to its source, while AI models increasingly rely on low-carbon energy grids and renewable resources. Leading tech companies have showcased significant emission reductions by utilizing energy-efficient data centers and advanced cooling technologies. For instance, applying DeepMind's machine learning to Google data centers, reduced the amount of energy for cooling by up to 40 percent according to the company. Salesforce trained its models in lower-carbon data centers, powered by electricity sources that emit nearly 70% less carbon than global average electricity. This resulted in 105 fewer tons of carbon dioxide equivalents than if data centers with global average carbon intensity were used for training models (Gamazaychikov, 2024; Walsh, 2024; Amazon Team, 2023; Intel, 2023; Thomas, 2022; Evans and Gao, 2016).

Moreover, AI can speed up renewable adoption, thereby reducing CO2 emissions. AI facilitates the integration of renewable energy sources by improving forecasts of the variability of solar or wind generation, thus balancing energy supply more effectively across the grid. Various AI-based approaches, including structured data management, data mining capabilities and machine learning techniques, support an ecosystem that enables smart energy grids (Rozite et al., 2023, Serebryantseva, 2023, and Şerban and Lytras, 2020). AI also supports the transition to greener energy sources by optimizing mining processes for critical materials and improving the efficiency of renewable energy infrastructure. Finally, *AI indirectly contributes to preserving nature* by speeding up the process of securing renewable energy supply (e.g. hydro).

What is the actual net balance of all these countervailing AI-energy-environment effects? Cross-country econometric analyses show that AI spending correlates positively with renewable energy adoption and reduced CO2 emissions, although its potential savings effect on energy consumption is less clear. This pattern is apparent in high-income and AI-leading countries, which would suggest the existence of a non–linear pattern linking energy consumption and AI spending.

Most research literature has focused so far on sustainable energy transition, consistently finding positive effects on greater adoption of renewable energy sources. Yin et al. (2023) analyze the impact of the number of AI software projects (as released in the GitHub platform) on the share of renewable energy as percent of total energy consumption for a sample of 62 economies covering the period 2011–2020. On this basis, the authors confirm a positive effect of AI software development on the sustainable energy transition in the following year, as it facilitates measuring and reporting consumption. Yang et al. (2024) confirm these results for a global sample of 44 countries from 2000 to 2022, estimating a positive effect of AI patents index on the share of renewable energy. Related to it, Chung and Hwang (2024) show that for 139 countries between 1990 and 2019, those that are proficient in both green and AI technology patents – especially when AI is integrated in green technologies - show greater capability in developing and sustaining green technological advancements. Finally, Rasheed et al. (2024) confirm this positive effect also on energy supply, based on the positive interaction of AI patents on renewable energy production in 22 leading robotics and innovative countries, from 1991 to 2020.

Regarding the impact on energy productivity (identified as energy efficiency, that is the lowest energy consumption required to achieve output), Hossin et al. (2023) show for 12 Middle East and North Africa (MENA) countries between 2000 and 2021, that the impact of AI (measured as applications to register industrial designs) consumption, is significant but weak. Katz and Jung (2024) showed that energy consumed by cloud computing and the information technology industry in general is more economically productive (higher value per unit of energy) than other sectors. More specifically in the United States, the states that have high deployment of cloud computing data centers and accelerate the migration to an IT-intensive economy benefitted from high energy productivity, measured as gross value added per unit of energy consumed (MWh).

In terms of the impact of AI on emissions, the evidence is mixed. For a sample of 14 East Asian and Pacific countries from 2000 to 2023, Sha et al. (2024) obtained different results regarding the impact of AI (measured by AI research publications) on CO2 emissions. Doran et al. (2024) showed inconclusive results on the impact of AI (measured by industrial robots, AI companies, and total and private AI investments in USD) on GHG emissions in 44 European countries during the period 2012-22. While the deployment of industrial robots and investment in AI technologies may have a significant impact on GHG emissions, and large companies exhibit higher investment levels potentially contributing to reduced emissions, authors claim the relationship between private investment in AI and GHG emissions requires further scrutiny.

Finally, Wang et al. (2024a) studied 67 countries between 1993 and 2019 and indicated that AI (proxied by the stock of industrial robots) significantly reduced the ecological footprint, reduced carbon emissions, and promoted renewable energies share. Furthermore, the environmental benefits of AI were more pronounced at higher levels of AI development. And Wang et al. (2024b) added to a similar sample of countries the role of trade openness, indicating that when trade openness crosses a certain threshold, AI has a significant negative impact on carbon emissions, and a positive impact of AI on energy transition. These last two papers - the most comprehensive studies similar to ours - suggested a non-linear effect between AI and environmental development in line with Lee and Yan (2024). These authors showed that the density of robots (as a proxy of AI) and the share of clean energy consumption in total energy consumption display a U-shaped relationship based on data covering Chinese provinces from 2016 to 2019.

Drawing on this literature, the main thesis of this paper is that AI will play a transformative role in creating a sustainable and greener future. While AI has in the combined increased energy consumption worldwide, it will also reduce the environmental footprint of economic growth and will speed up renewable energy's adoption. For that purpose, we will test the empirical relationship between AI and energy consumption, AI and energy productivity, AI and CO2 emissions, and AI and the share of renewable energy. We will check where there are potential non-linear relationships, as AI adoption might initially have negative environmental effects, more than compensated when it reaches widespread adoption across the economy. This opens a key global policy area since levels of AI spending are not affordable to or even available in low- and middle-income economies.

In sum, we contribute to this growing literature by first testing a complete set of AI effects: energy requirements, energy productivity, emissions and renewable energy adoption. Second, we explicitly contrast the non-linearity in these relationships. Third, we believe the AI market (measured as the acquisition of software and applications, hardware and business services to provide AI capabilities) is a more appropriate indicator to measure AI usage intensity compared to others used in the research literature (who rely on indicators such as the number of AI projects, AI publications, number of IP applications, or the number of industrial robots, number of AI companies). Finally, our sample covers most AI-leading countries starting in 2019 through 2023.

## 3. Dataset and descriptive analysis

We built a panel for 23 countries covering the period 2019-2023 of data on the AI market. The countries included in the panel represent 78% of the global GDP in 2024 according to the data reported by the International Monetary Fund (IMF), and practically all main AI users and producers worldwide (Table 1).

Table 1. **Countries covered in the database**

| | | |
|---|---|---|
| Argentina | Germany | Peru |
| Australia | Hong Kong[3] | Singapore |
| Brazil | India | South Africa |
| Canada | Israel | Spain |
| Chile | Italy | Turkey |
| China | Japan | United Kingdom |
| Colombia | Korea | United States |
| France | Mexico | |

[3] We treat Hong Kong separately from the PRC as it is usually considered a different entity in most statistical sources.

We define five dependent variables. First, we focus on energy required by end users (*energy consumption* hereafter) considering the total MW/h by country per year, measured by the quantity supplied from either local or foreign production.[4] Next, we include energy productivity, measured as the ratio GDP/Energy supplied (i.e. $ of output per MW/h). Third, we also consider dependent variables linked to emissions. For this purpose, we calculated the GDP/CO2 ratio (production per carbon dioxide emissions, measured as billion dollars per Mt) per country, and on overall emissions per capita (Mt of carbon dioxide emissions per person). Finally, we considered the national share of the total energy that is sourced from renewable sources (hydro, geothermal, solar, wind, or biofuels). Energy variables come from the IEA dataset, emissions from the World Bank (WB) dataset, and GDP from the IMF (Table 2).

Data on AI demand was compiled and validated from various sources including IDC, Gartner and Statista. Two important caveats are in order regarding our AI market definition. First, it captures AI use (that can potentially improve environmental outcomes), but development and training, which is energy and capital intensive and is less transparent might be less so. Second, it may not be properly capturing the whole environmental impact associated with its use if the datacenters are located in other countries. In this case, a part of the AI carbon footprint can be hidden from these statistics, outsourced to other countries and thus being complex to properly account for it. Both points may make our AI variable to underestimate the environmental impact associated with AI, thus being necessary to analyze the results with due caution.

Table 2. **Variables definitions**

| Group | Variable | Description | Mean | Std. Dev. | Source |
|---|---|---|---|---|---|
| Dependent variables | Energy pc | Total energy supply per capita (in MW/h). | 34.119 | 22.635 | IEA |
| | GDP / Energy | Production per unit of energy (dollars per MW/h) | 900.867 | 616.525 | IEA / IMF |
| | GDP / CO2 | Production per carbon dioxide emissions (billion dollars per Mt) | 4.771 | 2.706 | IMF / WB |
| | LCO2 pc | Logarithm of carbon dioxide emissions per capita (Mt) | -12.109 | 0.650 | WB |
| | Renewable share | Energy generated from hydro, geothermal, solar, wind, or biofuels (as % of total energy consumed) | 15.023 | 10.584 | IEA |

[4] *Total energy supply* (TES) includes all the energy produced in or imported to a country, minus that which is exported or stored. It represents all the energy required to supply end users in the country. Some of these energy sources are used directly while most are transformed into fuels or electricity for final consumption.

| | | | | | |
|---|---|---|---|---|---|
| Independent variable: AI | AI market per capita | AI market per inhabitant (in dollars) | 52.850 | 89.234 | Multiple sources |
| Control variables | Government spending | General government final consumption (% of GDP) | 17.306 | 4.014 | WB |
| | Openness | Trade (% of GDP) | 80.671 | 84.684 | WB |
| | Industry | Manufacturing, value added (% of GDP) | 13.984 | 6.208 | WB |
| | Education | Gross enrolment ratio, tertiary (%) | 74.602 | 24.188 | WB |
| | GDPpc | GDP per capita (in dollars) | 31490.74 | 22639.88 | IMF |

*Source:* Authors

As for control variables, we will include as it is usual in empirical literature, different indicators that can potentially explain energy and environmental patterns. First, we consider government spending (as a share of the GDP) as the public sector can potentially play a crucial role through actions of public interest involving significant energy consumption, such as buildings, transport, and public lighting (De Moraes e Soares et al., 2024). In addition, increased public-sector investment in energy transition may yield positive environmental outcomes. Second, we will also include openness (measured in terms of trade by GDP). As argued by Osei-Assibey Bonsu and Wang (2022), a link exists between openness and energy use, as international trade can be associated with increased economic activity, or eventually to potential variations in the composition of the national output.

Third, we include the weight of the manufacturing sector in the economy, as the sectoral structure is an important driver in this field in past research (Atalla and Bean, 2017; Sineviciene et al., 2017; Chang and Hu, 2010). In this sense, industrial activity can be associated with larger energy requirements than most service sectors. Education is also considered, as improved educational levels may change the energy consumption habits of people (Mahmood, 2020). Finally, we also include GDP per capita as a measure of development and standard of living, as several authors identified it as relevant to explain energy efficiency metrics (Atalla and Bean, 2017; Sineviciene et al., 2017; Song and Zheng, 2012; Jimenez and Mercado, 2014; Chang and Hu, 2010). All these control variables come from WB databases.[5]

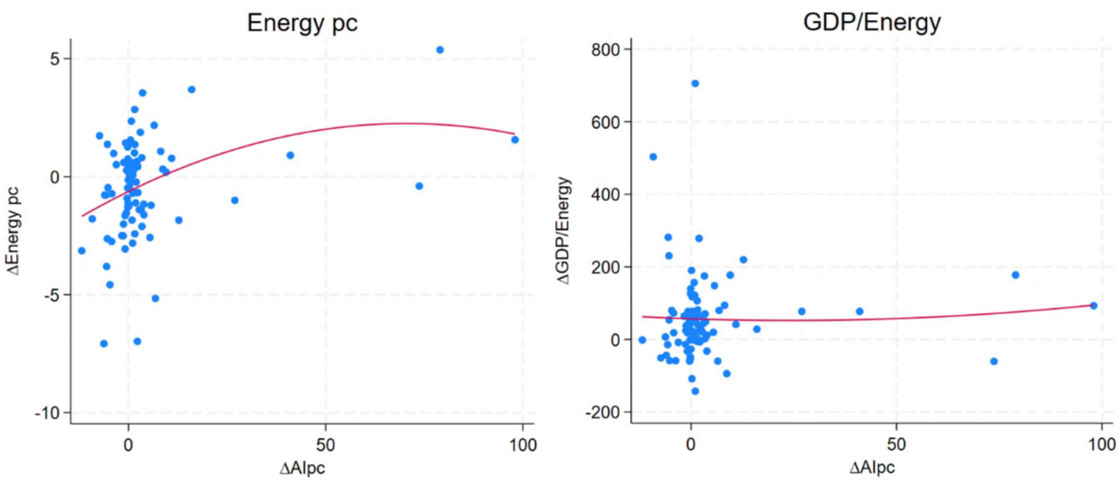
Preliminary evidence, based on the hypotheses described in Section 2, suggests a global increase in energy requirements due to the growth in AI spending. Additionally, a non-linearity pattern seems to emerge in the relationship between AI spending and energy requirements per capita, energy productivity and CO2 emissions, and to a lesser extent, the share of renewable energy. In all of these cases,

[5] We considered including further variables as controls, such as the degree of urbanization, geographic conditions, or local endowments of natural resources and energy sources. However, these indicators show limited variance over time, especially in the brief period of our panel, and their effects are likely absorbed by the country-level fixed effects.

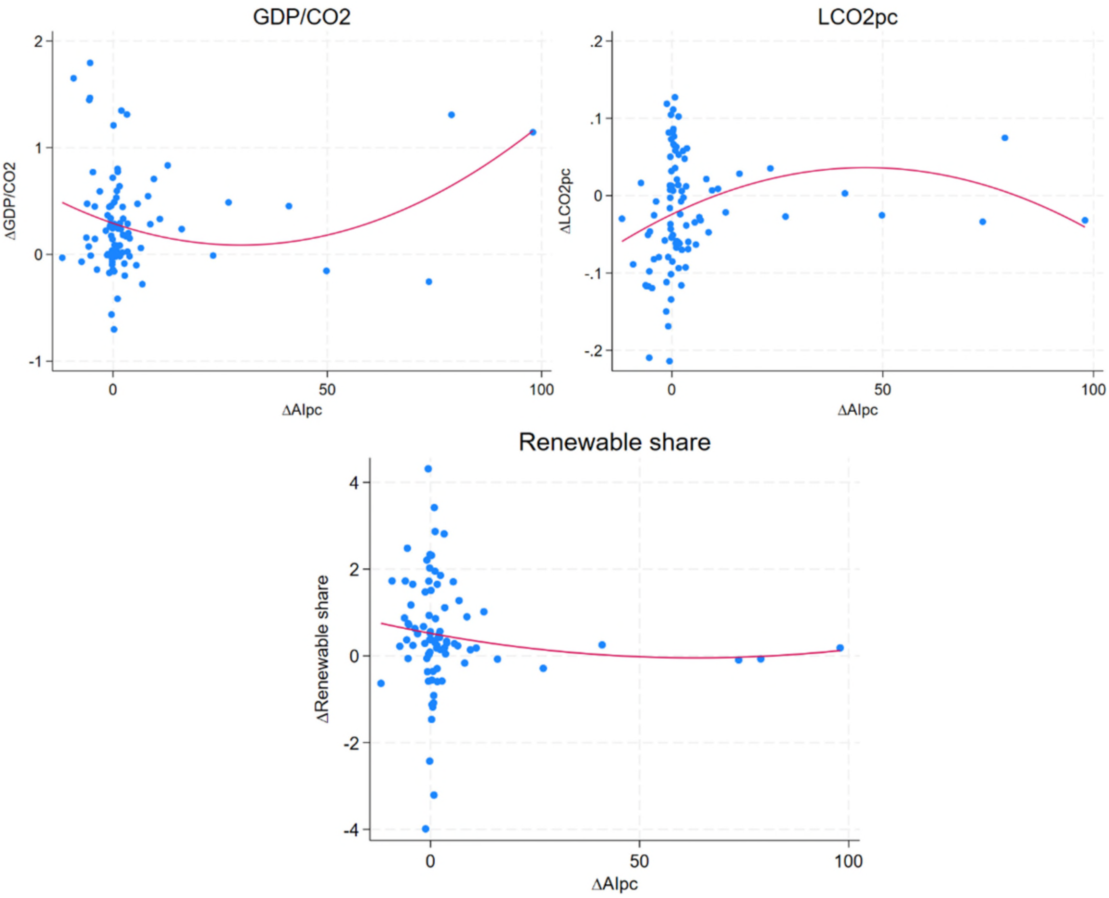
the visual evidence points to an initial negative effect from AI spending on the environment, which beyond a certain level of AI spending per capita turns out to be a positive one, reducing emissions (Figure 2). In other words, this suggest the existence of a *Kuznets curve* for green AI.[6]

In the next section, we will perform the econometric analysis to test the hypotheses described in Section 2 and assess whether these visual links between AI, energy and the environment hold empirically.

Figure 2. **Per capita AI spending** (USD) **and key energy and environmental indicators**




[6] The *Kuznets curve* refers to the hypothesis proposed by the economist Simon Kuznets in the 1950s-1950s showing that as economy develops (income per capita), market forces first thrive inequality up, and then down.




*Source:* Authors based on data from Statista, Gartner, IDC, IEA, WB and IMF

## 4. Econometric estimates

This section reports on the econometric results on the relationship between the AI market and energy and environmental performance for the sample of 23 countries, from 2019 to 2023. All estimates incorporate country and year fixed effects.[7]

### 4.1 AI and energy requirements

The first issue we address is whether AI increases energy consumption, or, conversely, if it is able to reduce it by stimulating efficiency gains. Initially, we present estimates that include the AI market regressor in levels and then introduce it in both levels and squares to account for potential non-linearities (Table 3).

[7] Country fixed effects absorb unobservable aspects related to cultural, idiosyncratic, or institutional reasons, if they are invariant over time. The addition of fixed effects per year, on the other hand, allows us to contemplate exogenous technological growth, as well as absorb any cyclical shocks that affect all economies. All estimates include robust standard errors.

**Table 3. AI market as driver of energy consumption and energy productivity**

| | (i) | (ii) | (iii) | (iv) | (v) | (vi) |
|---|---|---|---|---|---|---|
| Dep. variable: | Energy pc | Energy pc | GDP/Energy | GDP/Energy | Energy pc | GDP/Energy |
| AI market pc | 0.04542*** | 0.09461*** | -2.75782** | -4.53012*** | 0.08719* | -5.99490 |
| | [0.01025] | [0.02724] | [1.01138] | [1.57681] | [0.04855] | [5.00957] |
| AI market pc squared | | | | | 0.00417 | -0.00005 |
| | | | | | [0.00006] | [0.00530] |
| Gov. Spending | -0.40241 | -0.33222 | 5.74815 | 3.51524 | -0.45793* | 10.05124 |
| | [0.27000] | [0.23629] | [15.36017] | [18.40531] | [0.25700] | [14.39154] |
| Openness | -0.03428 | -0.11425* | 0.38094 | 3.18048 | -0.05449 | 1.94719 |
| | [0.05907] | [0.05875] | [3.27556] | [3.32461] | [0.05705] | [3.72558] |
| Industry | 0.35765 | 0.51683** | -36.16131* | -41.90103** | 0.34507 | -35.18644* |
| | [0.28263] | [0.25738] | [19.96805] | [19.44716] | [0.28689] | [19.75336] |
| education | 0.06531 | 0.04012 | 18.55466 | 19.78314*** | 0.06189 | 18.81979 |
| | [0.09751] | [0.06469] | [11.08666] | [6.86961] | [0.09372] | [11.01367] |
| GDP pc | -0.00013 | -0.00046*** | 0.03219*** | 0.04366*** | -0.00015 | 0.03381*** |
| | [0.00010] | [0.00014] | [0.00853] | [0.00861] | [0.00010] | [0.01034] |
| Under-id test | | 7.490** | | 7.490** | | |
| Weak-id test | | 12.731(γ) | | 12.731(γ) | | |
| Over-id test | | 0.148 | | 0.299 | | |
| Endogeneity test | | 7.479*** | | 3.700* | | |
| Country FE | YES | YES | YES | YES | YES | YES |
| Year FE | YES | YES | YES | YES | YES | YES |
| Observations | 97 | 91 | 97 | 91 | 97 | 97 |
| R-squared | 0.5170 | 0.4511 | 0.7376 | 0.7126 | 0.5236 | 0.7436 |
| Method | OLS | IV-LIML | OLS | IV-LIML | OLS | OLS |

*Note:* *** p<1, ** p<5%, * p<10%. Robust standard errors in brackets. (γ) Stock-Yogo weak ID test critical values: 10% maximal LIML size 8.68.
*Source:* Authors

The positive link between the AI market and energy supply per capita is clear. This result is verified in column (i) when using Ordinary Least Squares (OLS) and also in column (ii) when we make the estimation using Instrumental Variables (IV) to account for potential endogeneity in the link between AI and energy supplied.[8] This evidence is consistent with results reported in the case of energy productivity in columns (iii) and (iv). As energy consumption increases, the output per energy decreases, something that is verified again through both OLS and IV.

In columns (v) and (vi) we replicate the estimations for both energy outcomes introducing AI in both levels and squares, in order to check potential non-

[8] In the IV estimation, we use as instruments for AI spending the number of academic publications per capita on AI related topics by local universities with a temporal lag (published papers for periods 2000-2004 and 2005-2009). These instruments come from the *OECD AI Policy Observatory*. The contrasts conducted for the IV verify the suitability of the instruments, as under identification is rejected, the instruments are not weak, and the Hansen test of overidentification does not reject the null hypothesis of exogeneity. The significance of the endogeneity C-test indicator rejecting the null hypothesis provides support to the need of making this estimation through IV.

linearities.[9] Overall, we do not find evidence of non-linearities from the link of AI market and both Energy pc and GDP/Energy, as the squared AI regressor is never significant. As for the remaining controls, the results presented in Table 3 indicate that the sectoral structure of the economy is relevant, as the larger manufacturing sector becomes, the more energy intensive the overall economy is. The opposite effects seem to arise from GDP per capita, from where we can conclude that its increase is related to less energy requirements and its more efficient use.

In sum, we conclude that AI has effectively generated an increase in energy requirements, something that is verified both in per capita terms and in terms of output.

### 4.2 AI and emissions

Next, we estimate the drivers of emissions measured as the ratios GDP/CO2 and CO2 per capita (in this last case, the dependent variable is introduced in logs, due to better fit models). The main hypothesis is that AI can contribute to reducing emissions and/or increasing the output per emission. Results for both OLS and IV methodologies are presented in Table 4.

Our results confirm, in line with the previous findings of AI driving an increase in energy consumption, that AI also reduces the output per emission (column (i)) and increases overall CO2 per capita (column (iii)). These results are further verified when conducting the estimations through instrumental variable by controlling for endogeneity (using the same instruments as before for AI market per capita), in columns (ii) and (iv).[10]

However, the link between AI and the emissions-related variables may be non-linear. For that purpose, in columns (v) and (vi) we re-estimate the models for output per emission and CO2 per capita introducing the AI regressor both in levels and squares. Results are clear in pointing out that the link between AI and emission-related outcomes is non-linear, as in both cases the AI regressor is significant both in levels and in squares.

Moreover, we find that in the initial adoption phase the AI results in negative environmental effects, but once a certain threshold in AI usage is reached, the effect turns out to be positive. This is evident from the positive coefficient associated with the squared AI regressor in the GDP/CO2 estimation (for high values of AI spending, the technology drives an increase in output per emissions), as well as due to the

[9] These estimations had to be conducted only through OLS with fixed effects, due to the unavailability of a sound instrument for the squared AI regressor. However, this does not appear to be a serious concern, as up to now all IV estimations have provided support to the previous OLS findings.

[10] While the instruments again behave as expected, in this case there is less clarity on the need to treat AI as endogenous, as the C-test only rejects the null hypothesis of exogeneity in the estimation of GDP/CO2.

negative squared coefficient in the CO2 estimate, from where it can be said that for high values of AI spending, there is a reduction in emissions.

Table 4. **AI spending as driver of emissions and output per emission**

| Dep. variable: | (i) GDP/CO2 | (ii) GDP/CO2 | (iii) LCO2 pc | (iv) LCO2 pc | (v) GDP/CO2 | (vi) LCO2 pc |
|---|---|---|---|---|---|---|
| AI market pc | -0.00564** | -0.01493** | 0.00046** | 0.00163* | -0.03397** | 0.00437*** |
| | [0.00223] | [0.00750] | [0.00019] | [0.00089] | [0.01376] | [0.00090] |
| AI market pc squared | | | | | 0.00003** | -0.00001*** |
| | | | | | [0.00002] | [0.00000] |
| Gov. Spending | 0.03424 | 0.00609 | -0.02743*** | -0.02473*** | 0.07634 | -0.03323*** |
| | [0.04971] | [0.06736] | [0.00779] | [0.00919] | [0.04636] | [0.00797] |
| Openness | 0.00197 | 0.00228 | -0.00414* | -0.00407* | 0.02014* | -0.00665*** |
| | [0.00725] | [0.01676] | [0.00208] | [0.00244] | [0.01095] | [0.00168] |
| Industry | -0.04526 | -0.12642 | 0.00287 | 0.01389 | -0.0333 | 0.00122 |
| | [0.06045] | [0.09684] | [0.01133] | [0.01371] | [0.05741] | [0.01057] |
| Education | 0.03178 | 0.03932** | 0.00167 | 0.00093 | 0.03661* | 0.00101 |
| | [0.02413] | [0.01798] | [0.00293] | [0.00226] | [0.01852] | [0.00242] |
| GDP pc | 0.00014*** | 0.00020*** | -0.00001** | -0.00002** | 0.00017*** | -0.00001*** |
| | [0.00003] | [0.00004] | [0.00000] | [0.00001] | [0.00003] | [0.00000] |
| Under-id test | | 5.379* | | 5.379* | | |
| Weak-id test | | 6.330(Y) | | 6.330(Y) | | |
| Over-id test | | 0.048 | | 0.074 | | |
| Endogeneity test | | 5.227** | | 2.626 | | |
| Country FE | YES | YES | YES | YES | YES | YES |
| Year FE | YES | YES | YES | YES | YES | YES |
| Observations | 103 | 97 | 103 | 97 | 103 | 103 |
| R-squared | 0.8116 | 0.7174 | 0.4932 | 0.3613 | 0.8352 | 0.5384 |
| Method | OLS | IV-LIML | OLS | IV-LIML | OLS | OLS |

*Note*: *** p<1, ** p<5%, * p<10%. Robust standard errors in brackets. (Y) Stock-Yogo weak ID test critical values: 10% maximal LIML size 8.68.
*Source:* Authors'

As for the control variables, again it is verified that higher development (as measured through GDP per capita) is associated with lower emissions per person, or higher output per emission. There is also some evidence on the relevance of government spending and having more open and educated societies for improving the environmental outcomes related to emissions.

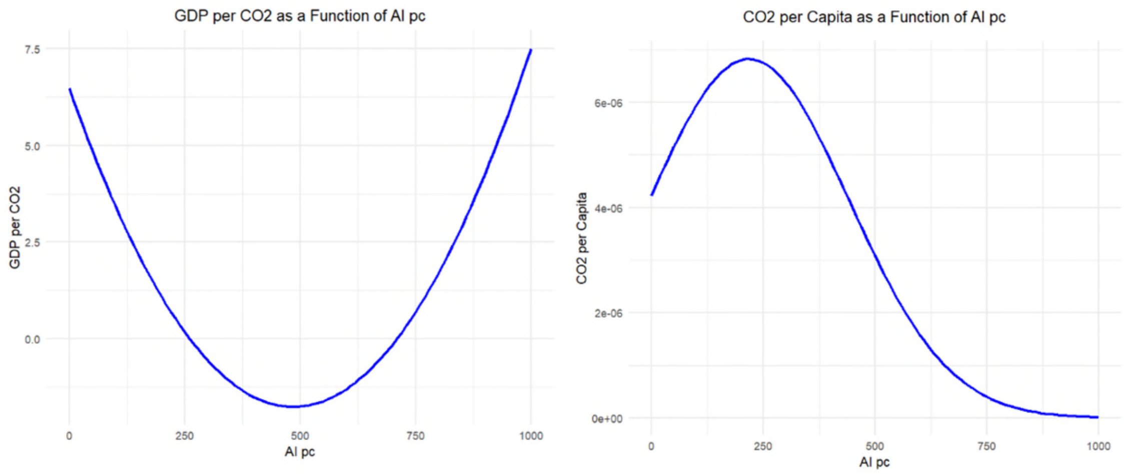
We then explore further these quantitative results to estimate the thresholds where AI market per capita has these positive environmental effects worldwide. For that goal, we use the coefficients estimated in Table 4 (including the constant term), and substituting for the sample average values of the variables, we get the respective functions linking emission outcomes with AI:

$$GDP/CO2 = 6.4741 - 0.03397 * (AI\ pc) + 0.00003 * (AI\ pc)^2$$

$$CO2\ pc = e^{[-12.37230+0.00437*(AI\ pc)-0.00001*(AI\ pc)^2]}$$

Where the CO2 pc has a different functional form as it has been un-logged by applying the exponential function (see Figure 3, for different AI pc levels). The relationship between AI and emissions is clearly non-linear. In the case of the output per emissions, the AI spending threshold after the technology starts yielding positive environmental outcomes is close to US$ 570 per population. This is a quite high AI spending level; currently only Singapore is close to it. As for the CO2 per capita estimation, the threshold is lower: at approximately US$ 220 per capita is when AI starts to contribute to emissions reduction. To date, only the United States and Singapore exhibit these spending figures. Overall, we can conclude that AI does not contribute to reducing CO2 emissions initially, although it does so beyond certain thresholds of AI spending.

**Figure 3. Estimated relationships between CO2 emissions and AI spending**




*Source:* Authors' own elaboration

A potential explanation of this pattern could exist in the AI spending pattern, for which we unfortunately lack relevant data. At lower levels of development, most spending of AI is dedicated to LLM training and foundational model development, as discussed in Section 2. Over time, as spending increases, two effects might be at work: AI is adopted in processes that result in growing energy efficiency, and AI use becomes less energy intensive since the increase in spending reflects a shifting mix from development to usage.[11]

[11] See Budaragina et al., (2024) for a view on the potential variance that exists between the so-called Global North/South.

### 4.3 AI and renewable energies

We finally test whether AI accelerates energy transition toward renewable energy sources. Results are presented in Table 5 for both OLS and IV approaches. The estimations that introduce the AI regressors in levels only indicate a negative coefficient associated with the spending in this technology. This effect is verified through both OLS and IV regressions presented in columns (i) and (ii), respectively (although the non-significant result for the endogeneity test suggests no need to use the IV approach in this case).

**Table 5. AI spending as driver of energy transition towards renewable energies**

| | (i) | (ii) | (iii) |
|---|---|---|---|
| Dep. variable: | Renewable share | Renewable share | Renewable share |
| AI market pc | -0.02131*** | -0.02621* | -0.05775** |
| | [0.00586] | [0.01543] | [0.02323] |
| AI market pc squared | | | 0.00005* |
| | | | [0.00003] |
| Gov. Spending | 0.38328 | 0.38031* | 0.43173* |
| | [0.22552] | [0.21247] | [0.21099] |
| Openness | 0.04060 | 0.04089 | 0.05824 |
| | [0.05080] | [0.05834] | [0.04952] |
| Industry | -0.04545 | -0.09888 | -0.03447 |
| | [0.17227] | [0.19360] | [0.17553] |
| Education | 0.09371 | 0.09812** | 0.0967 |
| | [0.06050] | [0.04085] | [0.05805] |
| GDP pc | 0.00019*** | 0.00022* | 0.00020*** |
| | [0.00007] | [0.00012] | [0.00007] |
| Under-id test | | 7.490** | |
| Weak-id test | | 12.731(Y) | |
| Over-id test | | 0.133 | |
| Endogeneity test | | 0.364 | |
| Country FE | YES | YES | YES |
| Year FE | YES | YES | YES |
| Observations | 97 | 91 | 97 |
| R-squared | 0.5659 | 0.5654 | 0.5750 |
| Method | OLS | IV-LIML | OLS |

*Note:* *** p<1, ** p<5%, * p<10%. Robust standard errors in brackets. (Y) Stock-Yogo weak ID test critical values: 10% maximal LIML size 8.68.
*Source:* Authors' own elaboration

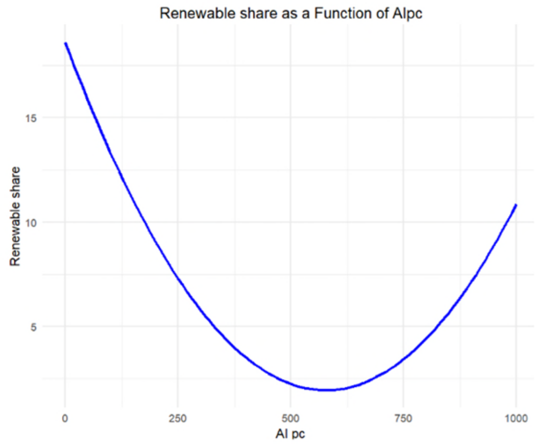
Once again, when testing non-linearities in column (iii), both coefficients for AI in levels and squares are statistically significant, and their signs suggest that initially AI reduces the share of renewables, but after a certain threshold the effect is reversed and turns positive. Again, GDP per capita is a relevant control, being in all cases significant to explain larger share of renewable energy. To a lesser extent, some evidence arises regarding the role of government spending and education. The reasons regarding why AI diffusion may drive an initial reduction of the renewables share are not clear. A possibility may be related to AI-driven efficiency gains in fossil-heavy grids. Alternatively, it may be that AI optimization does not necessarily prioritize renewable energy expansion unless explicitly guided by policy incentives. Our empirical analysis does not allow us to disentangle these hypotheses, meaning that this is an area that will require further investigation.[12]

Taking the coefficients estimated in Table 5, and substituting for the sample average values of the variables, we get the respective functions linking renewable share with AI:

$$Renewable\ share = 18.621 - 0.05775 * (AI\ pc) + 0.00005 * (AI\ pc)^2$$

The estimated function is plotted in Figure 4, for different AI market per capita levels. The share of renewable energy falls for initial AI spending levels, although it starts yielding a positive outcome from US$ 580 per population (in line with GDP/CO2 results). As mentioned before, only Singapore was close to this level in 2023.

**Figure 4. Estimated relationship between AI spending and renewable energies**




*Source:* Authors' own elaboration

[12] Additionally, we conducted estimates for non-renewable energy sources. The results indicate that AI spending conducts to higher share of non-renewables in general, and of coal in particular. In the estimation the same $ 580 dollars threshold of AI demand is the point from where the share of non-renewable energy starts to diminish. These results are available upon request.

### 4.4. Policy implications. Focus on low and middle-income countries

Overall, these results highlight the need for strategic vision and solid institutions to drive AI, both at the global and the national level. Indeed, at the global level there is technically some degree of governance, notably led by UN bodies, by the G20 and by the OECD. However, it is fragmented and divided, with a limited role for low and middle-income countries. This occurs despite the fact that natural resources – from rare earths, critical minerals and water – needed for batteries, semiconductors and data centers functioning are mostly located in low- and middle-income countries.[13]

The AI and energy agenda also requires national strategies, adapted to technological maturity and fiscal space. Using our AI market thresholds, a policy sequencing framework could start at early phases (e.g. <US$100 per capita of AI spending) focusing on establishing basic AI governance for energy-intensive sectors, launching low-emissions regulatory pilots, and supporting foundational R&D for local energy-AI startups. In a mid-phase (US$100 – $US300 per capita of AI spending) governments should accelerate domestic deployment of AI for energy system planning, foster sectoral innovation hubs, and embed green AI standards into public procurement frameworks. In an advanced-phase (>US$ 300 per capita of AI spending) countries could position national AI ecosystems as global leaders by scaling AI-driven clean energy innovation, exporting sustainable AI technologies, and embedding AI into long-term climate and industrial strategies.[14] In all stages, energy-rich countries could attract AI investment – smart data and computing centers run on renewable energies -, within productive development policies (Lebdioui et al., 2025).

On the financial front, achieving environmental benefits from AI is heavily dependent on quite high investments according to our estimates, which poses major inclusivity and equity challenges. This requires urgent action in the policy and finance environments if we are to accelerate towards an optimistic scenario. Conditioning public AI R&D support (e.g., tax incentives, procurement contracts, cloud service subsidies) on energy performance and emissions disclosure standards, and establishing public-private co-financing instruments targeting AI for energy access and distributed renewables (e.g. digital development bonds; Balmaseda et al., 2025[15])) should be explored.

---

[13] See Sirimanne and Fu (2025) for a focus on financial incentives to drive AI towards improving education, health and environmental sustainability, and Mazzucato and Ramos (2022) focus on its governance (regulation, institutions and skills).

[14] We would like Riad Meddeb, Director of the Sustainable Energy Hub at UNDP, for suggesting this point.

[15] Following the success of social and green bonds, *digital development bonds* fund projects such as fiber optic expansion, AI-driven smart cities, advanced cybersecurity frameworks, and sustainable data centers. These structured bonds would leverage asset-backed securities, where the

Finally, specific on AI data, transparency and standardization of measuring and reporting should be enhanced, to help shifting capital markets toward prioritizing low-emission, inclusive AI ecosystems. This could be done using mandatory carbon disclosures for AI development and data centers, or more market friendly instruments as enriched ESG frameworks and voluntary disclosures (e.g. *AI score* launched by Salesforce, Hugging Face, Cohere, and Carnegie Mellon University).

## 5. Conclusions and further research

AI global boom is currently putting pressure on the environment worldwide, given its intensive use of natural resources, minerals, water and energy. The speed and size of this impact has made the relationship between AI and the *green agenda* one of the economic, social and political key global topics.

By focusing on energy and environmental goals, AI will favor renewable energy transition and reduce emissions. While so far AI demand is significantly driving energy requirements and the CO2 footprint of countries and companies, this trend could shift over time, depicting significant variance across countries.

This paper presented novel empirical evidence on AI and its relationship with environmental indicators, based on an original database covering 23 AI-using and AI-producing countries. Our results confirm AI demand is driving significantly higher energy consumption in all countries. Also, due to this energy consumption, and the non-significant effect on renewable energies at lower spending levels, AI spending initially increases CO2 emissions. However, this does not necessarily imply that AI spending cannot foster the transition to sustainable energy. In fact, our empirical analysis also confirms the existence of certain non-linear relationships, that could be labeled as a *Kuznets curve* for green AI.

The positive effect on the environment in terms of emission reduction and higher reliance on renewable energies emerges beyond an AI market threshold; $580 per capita spending for renewable energy share; $570 for output per emissions and $220 in the case of CO2 per capita. Based on these thresholds, some countries would be already benefiting from this technological dividend. In particular, Singapore should be experiencing all positive effects, the United States is already reducing emissions due to AI, which at current AI spending per capita trends will reach the $570-$580 threshold in 2031. Korea will reach the $220 threshold in 2035 and $570-$580 in 2043. By contrast, the latest trends in European countries which

infrastructure itself serves as collateral. By linking financial returns to technological success, these bonds ensure long-term sustainability and reduced default risk. Governments and development banks could provide guarantees, ensuring projects meet long-term objectives and align with national digital strategies. Finally, as transparency is essential, they will require independent impact certification (Balmaseda et al, 2025).

depict extremely low growth rates of AI spending per capita anticipate that it will take several decades to reach positive continent-wide effects.

These results have quite clear policy implications. calling for a more coordinated global AI and energy governance, for national AI strategies adapted to the level of digital maturity, the need for innovative funding mechanisms (including the aforementioned *digital development bonds*) and enhanced transparency and standards measuring and reporting the use of energy by AI.

Our research faced some limitations stemming from data availability that prevented us from addressing some critical areas of analysis. First, our AI measure captures overall usage intensity, but not the specific demand for development and training of AI tools, which are highly energy intensive. Second, as a usage proxy, our AI variable may still not be fully capturing the environmental effect since the datacenters used in the process may be located abroad. Third, due to data aggregation we could not break down AI by application (e.g., energy, industry, agriculture, digital services). Finally, our panel extends through 2023 (again due to data availability not only in AI demand but also on energy consumption), which means we missed the latest wave of technological development, especially in terms of Generative AI.

Actually, after a brief slowdown, corporate investment in AI has rebounded, and newly funded generative AI startups nearly tripled in 2024 (Stanford Institute for Human-Centered AI, 2025). Its link to energy boomed, with some leading AI companies announcing the adoption of nuclear energy in US plants (pending regulatory approval and other technological issues) as a clean, sustainable power source for their data centers. Also, many countries have launched billion-dollar national AI infrastructure initiatives, including major efforts to expand energy capacity to support AI development.

As for potential research extension, in addition to expanding the sample as soon as more data becomes available, we would like to include a specific analysis on nuclear energy. Second, water consumption – needed both for semiconductors and batteries production and datacenters is an additional environmental impact dimension. Third, as energy has assumed strategic geopolitical implications, the empirical analysis of AI and energy could include security, covering both the reliability of the service and its privacy and cybersecurity dimensions given its speed digitization. Fourth, beyond decarbonization AI also supports inclusive development. Therefore, we could address the impact of AI on electricity coverage, on energy transmission and distribution losses (to proxy for grid efficiency and investment gaps), and on urban–rural access gaps. Finally, it would be interesting to work on a cost-benefit analysis to compare the cost per ton of emissions avoided through AI with the direct investment in renewables or energy infrastructure that may yield a similar effect.

## References


Acemoglu, D. (2024), "The Simple Macroeconomics of AI"; *NBER Working Papers* 32487. https://www.nber.org/papers/w32487

Amazon Team (2023), *Harnessing the power of plants to decarbonise our data centres.* https://www.aboutamazon.eu/news/sustainability/harnessing-the-power-of-plants-to-decarbonise-our-data-centres

Atalla, T. and P. Bean (2017), "Determinants of energy productivity in 39 countries: An empirical investigation", Energy *Economics*, 62, 217-229. https://www.sciencedirect.com/science/article/pii/S0140988316303498

Baily, M.N., E. Brynjolfsson and A. Korinek (2024), *Machines of mind: The case for an AI-powered productivity boom*. Brookings Research. https://www.brookings.edu/articles/machines-of-mind-the-case-for-an-ai-powered-productivity-boom/

Balmaseda, M., A. Melguizo and V. Muñoz (2025), *Financing the future: a proposal for digital development bond for emerging economies*. ARGIA and CCLATAM. https://www.cc-latam.org/financingthefuture-aproposalfordigitaldevelopmentbondsforemergingeconomies

Budaragina, M, D. West, A. Finotto Villena and A. Phillips (2024), *Powering AI in the Global South*. Tony Blauir Institute for Global Change. https://institute.global/insights/climate-and-energy/powering-ai-in-the-global-south

Chang, T.P. and J.L. Hu (2010), "Total-factor energy productivity growth, technical progress, and efficiency change: An empirical study of China", Applied *Energy*, 87(10), 3262-3270. https://www.sciencedirect.com/science/article/pii/S0306261910001339?casa_token=kieqVhust54AAAAA:DLIjgGw2kWRYtObWkJuUk-mdcjJSmtbi6dx05qbFGJr_VNIYyDY0hvLVqx8o_p6nbsfrI4h6yg

Chun, Y., and J. Hwang (2024), "The Nexus of Artificial Intelligence and Green Innovation: A Cross-Density Analysis at the Country Level." *Journal of the Knowledge Economy* https://doi.org/10.1007/s13132-024-02076-8

Crownhart, C. (2024), "*AI is an energy hog. This is what it means for climate change",* *MIT Technology Review.* www.technologyreview.com/2024/05/23/1092777/ai-is-an-energy-hog-this-is-what-it-means-for-climate-change

De Moraes e Soares, R.., A.M. Nunes, O. Pinheiro, A.C. Kaizeler and V. Martins (2024), "Asymmetries in energy consumption: efficiency of public spending across Portuguese municipalities", *Public and Municipal Finance*, 13(2), 110-128. https://www.businessperspectives.org/index.php/journals/public-and-municipal-finance/issue-466/asymmetries-in-energy-consumption-efficiency-of-public-spending-across-portuguese-municipalities

Doran, N.M., G. Badareu, M.D. Doran, M. Enescu, A.L. Staicu , and M. Niculescu (2024), "Greening Automation: Policy Recommendations for Sustainable Development in AI-Driven Industries." *Sustainability*, 16(12), 4930. https://doi.org/10.3390/su16124930

Evans. R. and G. Gao (2016), *DeepMind AI Reduces Google Data Centre Cooling Bill by 40%.* https://deepmind.google/discover/blog/deepmind-ai-reduces-google-data-centre-cooling-bill-by-40/

Gamazaychikov, B. (2024), *Unveiling Salesforce's Blueprint for Sustainable AI: Where Responsibility Meets Innovation.* Salesforce Engineering. https://engineering.salesforce.com/unveiling-salesforces-blueprint-for-sustainable-ai-where-responsibility-meets-innovation/?_ga=2.166247287.877827697.1730276307-560183023.1723741026

Hossin, M.A., D. Alemzero, R. Wang, M. Kamruzzaman, and M.N. Mhlanga (2023), "Examining Artificial Intelligence and Energy Efficiency in the MENA Region: The Dual Approach of DEA and SFA." *Energy Reports* 9 (2023) 4984–4994. https://doi.org/10.1016/j.egyr.2023.03.113

Intel (2023), *Intel Dives into the Future of Cooling.* https://www.intel.com/content/www/us/en/newsroom/news/intel-dives-into-future-of-cooling.html

International Energy Agency (2025), *Energy and AI. World Energy Outlook Special Report*. https://www.iea.org/reports/energy-and-ai/energy-demand-from-ai

ITU (2024), *AI and the Environment - International Standards for AI and the Environment. 2024 Report.* International Telecommunication Union. https://www.itu.int/dms_pub/itu-t/opb/env/T-ENV-ENV-2024-1-PDF-E.pdf

ITU and WBA (2024), *Greening Digital Companies 2024.* https://www.itu.int/en/ITU-D/Environment/Pages/Publications/GDC-24.aspx. International Telecommunication Union and the World Benchmarking Alliance.

Jimenez, R. and J. Mercado (2014), "Energy intensity: A decomposition and counterfactual exercise for Latin American countries", Energy *Economics*, 42, 161-171. https://www.sciencedirect.com/science/article/pii/S0140988313002983?casa_token=QIqzrd4OPyAAAAAA:vjWaK6dEozaIiR8fMN-XWJsU4aNAF5z3yrWagcCK81JkwVW8RIvIcSKnsOUuK4MimZwjwlKw7A

Katz, R. and Jung, J. (2024). *Comparative assessment of energy productivity of cloud computing and information technologies in the United States*. New York: Telecom Advisory Services. https://www.teleadvs.com/comparative-assessment-of-energy-productivity-of-cloud-computing-and-information-technology-in-the-united-states

Lebdioui, A., A. Melguizo and V. Muñoz (2025), "Artificial Intelligence, Biodiversity & Energy: From a Resource-Intensive to a Symbiotic Tech". Oxford University, TIDE Centre *working paper* 90. https://oxford-tide.org/wp-content/uploads/2025/01/tide-working-paper-90_-ai-biodiversity-and-energy5.pdf

Lee, C.C. and J. Yan (2024), "Will artificial intelligence make energy cleaner? Evidence of nonlinearity," *Applied Energy*, 363. www.sciencedirect.com/science/article/abs/pii/S0306261924004641

Lewis, D. and J. Oxby (2024), "Energy security and AI", *UK Parliament POST*. https://researchbriefings.files.parliament.uk/documents/POST-PN-0735/POST-PN-0735.pdf

Lokwon, K. (2025), *How on-device AI can help us cut AI's energy demand*, World Economic Forum Centre for the Fourth Industrial Revolution. https://www.weforum.org/stories/2025/03/on-device-ai-energy-system-chatgpt-grok-deepx/

Luccioni, A.S., Y. Jernite and E. Strubell (2024), "Power Hungry Processing: Watts Driving the Cost of AI Deployment?" *ACM Conference on Fairness, Accountability, and Transparency.* https://arxiv.org/abs/2311.16863

Mahmood, H. (2020), "Level of education and renewable energy consumption nexus in Saudi Arabia", MPRA *working paper* 109141. https://mpra.ub.uni-muenchen.de/109141/1/MPRA_paper_109141.pdf

McKinsey (2023), *The economic potential of generative AI: The next productivity frontier.* https://www.mckinsey.com/capabilities/mckinsey-digital/our-insights/the-economic-potential-of-generative-ai-the-next-productivity-frontier

Mehta, S. (2024), *How Much Energy Do LLMs Consume? Unveiling the Power Behind AI.* https://adasci.org/how-much-energy-do-llms-consume-unveiling-the-power-behind-ai/

OECD (2024), "Miracle or Myth? Assessing the macroeconomic productivity gains from Artificial Intelligence", OECD *Artificial Intelligence Papers.* https://www.oecd.org/en/publications/miracle-or-myth-assessing-the-macroeconomic-productivity-gains-from-artificial-intelligence_b524a072-en.html

Osei-Assibey Bonsu, M., and Y. Wang (2022), "The triangular relationship between energy consumption, trade openness and economic growth: new empirical evidence", Cogent *Economics & Finance*, 10(1), 2140520. https://www.tandfonline.com/doi/full/10.1080/23322039.2022.2140520

Ramos, G. and M. Mazzucato (2022), "AI in the Common Interest", *Project Syndicate*. https://www.project-syndicate.org/commentary/ethical-ai-requires-state-regulatory-frameworks-capacity-building-by-gabriela-ramos-and-mariana-mazzucato-2022-12

Rasheed, M.Q., Y. Zhao, Z. Ahmed, A. Haseeb, and S. Saud (2024), "Information Communication Technology, Economic Growth, Natural Resources, and Renewable

Energy Production: Evaluating the Asymmetric and Symmetric Impacts of Artificial Intelligence in Robotics and Innovative Economies." *Journal of Cleaner Production*, 447: 141466. https://doi.org/10.1016/j.jclepro.2024.141466

Rojek, I., A. Mroziński, P. Kotlarz, M. Macko and D. Mikołajewski (2023), *AI-Based Computational Model in Sustainable Transformation of Energy Markets*. https://www.mdpi.com/1996-1073/16/24/8059

Rosalsky, G. (2025), *Why the AI world is suddenly obsessed with Jevons paradox : Planet Money.* NPR. https://www.npr.org/sections/planet-money/2025/02/04/g-s1-46018/ai-deepseek-economics-jevons-paradox

Rozite, V., J. Miller and S. Oh (2023), "Why AI and energy are the new power couple," *IEA Commentary.* https://www.iea.org/commentaries/why-ai-and-energy-are-the-new-power-couple

Şerban, A.C. and M. D. Lytras (2020), "Artificial Intelligence for Smart Renewable Energy Sector in Europe—Smart Energy Infrastructures for Next Generation Smart Cities," *IEEE Access*, 8, 77364-77377 https://ieeexplore.ieee.org/document/9076660

Serebryantseva, V. (2023), *AI in the Energy Industry: Impact, Benefits, and Use Cases.* https://pixelplex.io/blog/ai-energy-industry/

Shah, S.A.; X. Ye, B. Wang, X. Wu (2024), "Dynamic Linkages among Carbon Emissions, Artificial Intelligence, Economic Policy Uncertainty, and Renewable Energy Consumption: Evidence from East Asia and Pacific Countries". *Energies* 17, 4011. https://doi.org/10.3390/en17164011

Sineviciene, L., I. Sotnyk and O. Kubatko (2017), "Determinants of energy efficiency and energy consumption of Eastern Europe post-communist economies", Energy *and Environment*, 28(8), 870-884. https://journals.sagepub.com/doi/full/10.1177/0958305X17734386?casa_token=NvnazHx4KSQAAAAA%3AeDpyimGiIDGJnJmxlV-Sn7e1W8E52zGqZXnCGHrv0KC1e7YfB5MXhWPtmJdrjTapjvYqbWAvTh_O

Sirimanne, S. and X. Fu (2025), "Will AI Close or Widen theDevelopment Gap?", Project Syndicate. https://www.project-syndicate.org/commentary/ai-wont-help-developing-countries-unless-governments-civil-society-step-in-by-shamika-sirimanne-and-xiaolan-fu-2025-04

Song, F. and X. Zheng (2012), "What drives the change in China's energy intensity: combining decomposition analysis and econometric analysis at the provincial level", Energy *policy*, 51, 445-453. https://www.sciencedirect.com/science/article/pii/S0301421512007264?casa_token=IAMNn353Ge4AAAAA:yfYp-__2y1gRw4Gx01dD4NROKKGSVso_O1GJh1GP9opvf0lTctGWYgCbD3jmUK_CXsNa419eHw

Stanford Institute for Human-Centered AI (2025), *Artificial Intelligence Index Report 2025*. https://hai-production.s3.amazonaws.com/files/hai_ai_index_report_2025.pdf

Stanford Institute for Human-Centered AI (2024), *Artificial Intelligence Index Report 2024.* aiindex.stanford.edu/wp-content/uploads/2024/05/HAI_AI-Index-Report-2024.pdf

Stanford Institute for Human-Centered AI (2023), *Artificial Intelligence Index Report 2023.* HAI_AI-Index-Report_2023.pdf

Tace, Y., S. Elfilali, M. Tabaa and C. Leghris (2023), "Implementation of smart irrigation using IoT and Artificial Intelligence," *Mathematical Modelling and Computing*, 10 (2), 575–582. https://science.lpnu.ua/mmc/all-volumes-and-issues/volume-10-number-2-2023/implementation-smart-irrigation-using-iot-and

Walsh, N. (2024), *Sustainable by design: Transforming datacenter water efficiency.* The Microsoft Cloud. https://www.microsoft.com/en-us/microsoft-cloud/blog/2024/07/25/sustainable-by-design-transforming-datacenter-water-efficiency/

Wang, Q., Y. Li, and R. Li (2024a). "Ecological Footprints, Carbon Emissions, and Energy Transitions: The Impact of Artificial Intelligence (AI)." *Humanities and Social Sciences Communications* 11:1043. https://doi.org/10.1057/s41599-024-03520-5

Wang, Q., F. Zhang, R. Li, and J. Sun (2024b), "Does Artificial Intelligence Facilitate the Energy Transition and Curb Carbon Emissions? The Role of Trade Openness." *Journal of Cleaner Production*, 447: 141298. https://doi.org/10.1016/j.jclepro.2024.141298

Yang, S., J. Wang, K. Dong, X. Dong, K. Wang, and X. Fu (2024), "Is Artificial Intelligence Technology Innovation a Recipe for Low-Carbon Energy Transition? A Global Perspective." *Energy* 300: 131539. https://doi.org/10.1016/j.energy.2024.131539

Yin, HT, J. Wen, and CP Chang (2023), "Going Green with Artificial Intelligence: The Path of Technological Change Towards the Renewable Energy Transition." *Oeconomia Copernicana* 14(4), 1059–1095. https://doi.org/10.24136/oc.2023.032